Last time we talked about comparing models and deciding which predictor variables to include.

In multiple linear regression, it is also still important to check whether the model assumptions appear to be satisfied in the data we're analyzing.

Most of the assumptions for the multiple linear regression model are the same as in the simple linear regression model:

(1) * linearity (of true relationship)

error terms are:
(2) * mean zero ← don't need to check this
(3) * independent
(4) * constant variance
(5) * normally distributed

Like before, the assumption about normality of the error terms is only required for inference (confidence intervals, hypothesis tests). The least squares estimates of the coefficients are valid without 3,4,5.

Also, it's still true that we don't need to check the assumption that the error terms have mean zero.

In the multiple linear regression model, we can assess linearity by looking at scatterplots of each individual predictor and the response, or each predictor and the residuals.

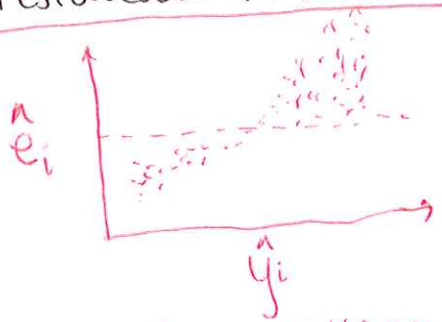


scatterplot for each predictor x_k

If any of these plots show a nonlinear pattern, that

indicates a violation of assumptions. You can consider trying a non-linear transformation of that predictor. Problems with nonconstant error variance may also show up in these plots.

It can also be helpful to look at a plot of the residuals ($\hat{e}$) vs. the fitted values $\hat{y}$. The residuals should be scattered evenly above and below zero, with no obvious pattern.



good: no pattern
bad: pattern
try transforming y

patterns in this plot can indicate violations of linearity, and/or problems with nonconstant error variance. Transforming predictors and/or the response variable y may help.

There is one new potential problem we need to consider now that there are multiple predictors: collinearity.

Collinearity refers to a scenario in which one of the predictors is equal to a linear combination of some of the other predictors. Conceptually, this is an issue when two or more predictor variables are closely related to each other.

If two predictor variables are highly correlated with each other, it can cause problems with the model. Let's start by considering two predictors that are perfectly correlated.

Suppose we want to model a person's height, based on the length of their forearm. We can fit a simple linear regression model and obtain the best-fitting line. Then imagine we also include a second predictor: forearm length, measured in meters instead of cm.

Y = height, cm

X_1 = arm length, cm

$\hat{Y} = 37 + 5X_1$ best fitting line

X_2 = arm length, m

The sample correlation between X_1 and X_2 will equal 1. If we know someone's forearm length in cm, we also know their exact forearm length in meters.

$$X_2 = \frac{X_1}{100} \qquad \text{correl}(X_1, X_2) = 1$$

Fit model: $\hat{Y} = \hat{\beta}_0 + \hat{\beta}_1 X_1 + \hat{\beta}_2 X_2 = \hat{\beta}_0 + \hat{\beta}_1 X_1 + \hat{\beta}_2 \frac{X_1}{100}$

$$= \hat{\beta}_0 + X_1\left(\hat{\beta}_1 + \frac{\hat{\beta}_2}{100}\right)$$

Our original "best-fitting" line with only arm length in cm as a predictor was $\hat{Y} = 37 + 5X_1$. Now, with X_1 and X_2 as predictors, there are multiple choices for $\hat{\beta}_1$ and $\hat{\beta}_2$ that effectively give us that same regression line.

A: $\hat{Y} = 37 + 5X_1 + 0X_2$

B: $\hat{Y} = 37 + 0X_1 + 0.05X_2$

C: $\hat{Y} = 37 + 2.5X_1 + 0.025X_2$

} all effectively same

For any arm length we could plug in, any of those models would give exactly the same predicted height.

A, B, C : same $\hat{Y}$ for any arm length

Because those models make exactly the same prediction for any arm length, the residuals of the fitted models are also the same, and the sum of squared residuals is also identical for all 3 models.

same $\hat{Y}$'s → same resids $(Y_i - \hat{Y}_i)$

↳ same $\sum_{i=1}^{n} (Y_i - \hat{Y}_i)^2$

[note: these are just 3 examples. Any model with $\hat{\beta}_1 + \frac{\hat{\beta}_2}{100} = 5$ is also the same as these 3.]

So if model A minimizes the sum of squared residuals, models B and C must also achieve the minimum sum of squared residuals. Thus, there are multiple choices of coefficients $(\hat{\beta}_1, \hat{\beta}_2)$ that all give the "best" model.

↳ multiple "best" models

If we have perfectly correlated predictors, then we cannot use the method of least squares to fit a linear regression model, because there is more than one set of "best" coefficients that give the minimum sum of squared residuals.

if correl$(X_1, X_2) = \pm 1$,
can't use least squares method to choose $\hat{\beta}_1, \hat{\beta}_2$

In practice, it is very unlikely that the observed predictor variables will have a correlation exactly equal to 1 or -1. As long as the correlation is not exactly ± 1 we can use least squares to fit a linear regression model. But a correlation very close to ± 1 can still cause problems.

$\text{corr}(X_j, X_k) = \pm 1$ can't use least squares

$\text{corr}(X_j, X_k)$ close to ± 1 : can use least squares but may have problems

To illustrate the problems we might face, imagine in the arm length/height data, we have measurements for each

person's left forearm (x_L) and right forearm (x_R), and we want to use BOTH measurements to predict height.

y = height
x_L = left arm length
x_R = right arm length

$\hat{y} = \hat{\beta}_0 + \hat{\beta}_1 x_L$ $\quad$ $\hat{y} = \tilde{\beta}_0 + \tilde{\beta}_1 x_L + \tilde{\beta}_2 x_R$

The left and right forearm lengths aren't identical, but they are highly correlated. Suppose the correlation is 0.99.

$\text{corr}(x_L, x_R) = 0.99$

Because we know the best fitting line using a single arm length was $\hat{y} = 14.8 + 5 * \text{arm length}$, and the left and right arm lengths are ALMOST the same, any model that has $\tilde{b}_1 + \tilde{b}_2 \approx 5$ will be very close to the best fit.

best fit: $\hat{y} = 37 + 5x_L$

in MLR model, if $\tilde{\beta}_1 + \tilde{\beta}_2 \approx 5$, then predictions $\hat{y}$ will be almost the same, because $x_L \approx x_R$

Whether we have $\tilde{\beta}_1 = 5, \tilde{\beta}_2 = 0$ or $\tilde{\beta}_1 = 2.5, \tilde{\beta}_2 = 2.5$

$\tilde{\beta}_1 = 0, \tilde{\beta}_2 = 5$, the resulting model for y is almost exactly the same.

→ all give very similar $\sum_{i=1}^{n} (y_i - \hat{y}_i)^2$

Because the correlation of x_L and x_R isn't exactly ± 1, there is one unique "best-fitting line for our observed data set.

because $\text{corr}(x_L, x_R) \neq \pm 1$, there IS a best solution $\tilde{\beta}_1, \tilde{\beta}_2$

But which values of the coefficients are "best" depends mostly on randomness in the observed data set. If we went out and collected a new set of sample data, the "best" coefficients could be very different, because the sum of squared residuals will be similar for any pair $(\tilde{\beta}_1, \tilde{\beta}_2)$ that satisfy $\tilde{\beta}_1 + \tilde{\beta}_2 \approx 5$.

"best" could be $\tilde{\beta}_1 = 4, \tilde{\beta}_2 = 1$

collect NEW sample data

"best" fit for new data: $\tilde{\beta}_1 = 8.1, \tilde{\beta}_2 = -3$

collect another new data set: $\tilde{\beta}_1 = 2.7, \tilde{\beta}_2 = 1.3$

Small changes in the data can produce big changes in the estimated coefficients, because there are many choices that give almost the same sum of squared residuals.

when predictors are highly correlated,
estimated coefficients can vary a lot
for sample datasets from same population

When predictors are highly correlated, it can be difficult to look at data and untangle what relationship each individual predictor has with the response. The estimates of the coefficients have more uncertainty. In other words, the estimates have greater variability.

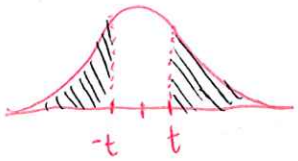
large $\text{corr}(X_j, X_k)$ → large variance & std error for $\hat{\beta}_j, \hat{\beta}_k$.

When we fit a linear regression model with highly correlated predictors, we aren't very confident in our estimates of their slope coefficients, and it is harder to determine whether those slopes are truly different from zero.

to test $H_0: \beta_k = 0$, test statistic is $\dfrac{\hat{\beta}_k}{\text{std error}(\hat{\beta}_k)} = t$

large std error → test stat close to 0

→ large pvalue, not significant



If the slope coefficient is truly nonzero, $\beta_k \neq 0$, then highly correlated predictors make us less likely to reject $H_0: \beta_k = 0$ (which is false, so we want to reject).

[Note: the probability of correctly rejecting the null hypothesis, when it is false, is called the "power" of a hypothesis test).]

For this reason, if we have highly correlated predictors in our model, we might choose to remove one of them.

remedy: remove one of the correlated predictors

To check whether any of the predictors are correlated, we can look at scatter plots of each pair of predictors.

We can also compute the sample correlation for each pair of predictors.

Collinearity can also involve more than 2 predictors – in which case it is often called "multicollinearity".

correlation among more than 2 predictors

This problem arises when one of the predictors is a linear combination of other predictors.

$X_k = 5X_j + 2X_m$ linear combination of predictors

One example would be if we wanted to predict college students' GPAs using their SAT scores, and we used 3 predictors: SAT math score, SAT reading score, and total SAT score.

y = GPA
X_1 = SAT math
X_2 = SAT reading
X_3 = SAT total

$X_3 = X_1 + X_2$

We cannot fit a linear model with those predictors using the method of least squares.

can't use least squares – best coefficients not unique

Once again, having a redundant variable in the model results in there being multiple sets of coefficients that all give the same minimum sum of squared residuals, because they make exactly the same predictions $\hat{y}$.

When more than 2 predictors are involved, this issue is harder to detect. If multicollinearity is exact, then when we try to fit the model, it just won't work. But just like with pairwise correlations, having ALMOST exact multicollinearity still makes the estimated coefficients have high variance.

exact collinearity → can't fit model
almost exact → can fit model, but high variance for coefficients
↪ harder to detect

When the relationship is among more than 2 predictors, the correlation between any pair of them might not be close to ±1

if $X_3 = X_1 + X_2$, $\text{corr}(X_1, X_3)$ and $\text{corr}(X_2, X_3)$ won't always be close to ±1

So looking at each pair of predictors won't always reveal this issue. A diagnostic quantity we can check instead is the "variance inflation factor".

checking pairwise correlations not enough
instead: "variance inflation factor" (VIF)

If predictor X_k is closely related to several other predictors in the model, then we aren't able to estimate β_k very precisely. The estimate has high variance.

A useful question to ask is: how much higher is the variance of the estimate $\hat{\beta}_k$ in the multiple linear regression model, compared to a simple linear regression using only predictor x_k?

2 models: $\hat{y} = \hat{\beta}_0 + \hat{\beta}_k x_k$ $\quad$ $\hat{y} = \tilde{\beta}_0 + \tilde{\beta}_1 x_1 + \ldots + \tilde{\beta}_k x_k$

→ how much bigger is $\text{var}(\tilde{\beta}_k)$ than $\text{var}(\hat{\beta}_k)$?

If the predictors in the multiple regression model have multicollinearity issues, then we aren't able to estimate the coefficient for x_k as precisely anymore, when including those other predictors.

→ if MUCH bigger, indicates multicollinearity involving x_k

this quantity, $\frac{\text{var}(\tilde{\beta}_k)}{\text{var}(\hat{\beta}_k)}$ is the "VIF"

you are not required to know the formula. We will not calculate this from scratch.

VIF ≥ 1 always

VIF > 5 or 10 indicates possible problem

There is no exact cutoff, but a value larger than 10 or so suggests multicollinearity issues. You can try removing predictors from the model.

remedy: can remove predictors

(but if collinearity isn't exact, the model is still valid)

If the multicollinearity is not exact, ie one predictor is not EXACTLY equal to a linear combination of the others, then the assumption is not violated. You can still fit the model and perform inference.

But you may choose to remove predictors to make the model more useful.

removing predictors may make model more useful

Which predictors to include is ultimately the data analyst's choice.

Note that this discussion of collinearity is specific to linear associations between predictors. Including two (or more) predictors that have a strong nonlinear association is not a problem in this regard.

nonlinear assoc between X_1 & X_2 is fine

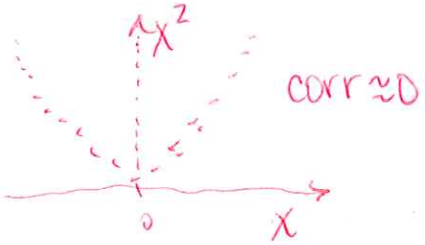
For example, we considered modeling tide depth as a function of time of day:

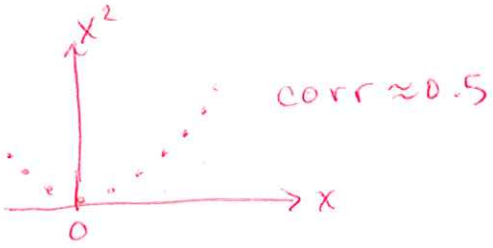
y = tide depth
x_1 = time of day

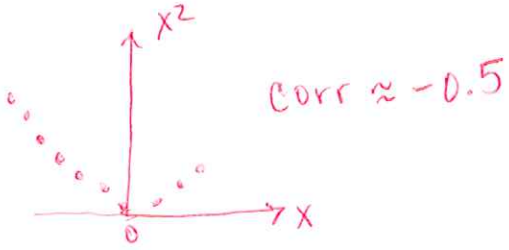
$\hat{y} = \hat{\beta}_0 + \hat{\beta}_1 x_1 + \hat{\beta}_2 x_1^2$

$corr(x_1, x_1^2)$ not always large

There is a strong nonlinear relationship between x_1 and x_1^2, of course, but the correlation coefficient describing their linear association can be very small (depending on the range of x_1 values included).







More generally, including polynomial terms x^2, x^3, x^4, etc. does not necessarily create problems with multicollinearity.

Including polynomial terms, eg $x^2, x^3, x^4, \ldots$ as predictors
- not necessarily collinearity problem

But adding these terms does come with the same issue as adding more predictors to the model in general - making the model more complicated/flexible always gives a better fit to the observed sample data, whether or not it ACTUALLY does a better job modeling the true population relationship.

- but adding more predictors always improves fit to SAMPLE data, even if they don't give closer match to TRUE model
  ↳ can be misleading!

So typically, we consider adding polynomial terms when subject matter knowledge tells us there SHOULD be a polynomial relationship, or when diagnostic plots show a specific nonlinear pattern. But we don't usually add polynomial terms arbitrarily just to see what happens.